Lec-4

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① Number System ⑪ Data Type ⑪ Type Casting

① Binary number System → 2 → [0, 1]
⑪ decimal number System → 10 → [0, 9]
⑪ octal number System → 8 → [0, 7]
→ 5 → [0, 4]

01101
10110 1
+ 00110 1
11101 0

2%2=0
2/2=1
1%2=1
1/2=0
3%2=1
3/2=1

decimal
$378 = 3\times10^2 + 7\times10^1 + 8\times10^0$

1 1
3784
+ 2687
6471

11%10=1 6%10=0
11/10=1 6/10=0
17%10=9
17/10=1
14%10=4
14/10=1

1 99
99
198

1 1 1
2345
4321
1110

6%6=0
6/6=1
7%6=1
7/6=1

:1101

2	39
2	19
2	9
2	4
2	2
2	1
	0

1×10^0
1×10^1
1×10^2
0×10^3
0×10^4
1×10^5

100111 → $100111 = 1\times10^5 + 0\times10^4 + 0\times10^3$
$1\times10^2 + 1\times10^1 + 1\times10^0$

while(n>0) {
rem = n % 2
sum = sum + rem × mul
n = n/2
mul = mul×10
}

sum = $1\times10^0 + 1\times10^1 + 1\times10^2 + 0 + 0 + 10^5$

mul = 1 10 10^2 10^3 10^4 10^5

2	39
2	19
2	9
2	4
2	2
2	1
	0

1×10^0
1×10^1
1×10^2
0×10^3
0×10^4
1×10^5

100000
100
10
100111

1+4+32 = 37

10	100101
10	10010
10	1001
10	100
10	10
10	1
	0

1×2^0
0×2^1
1×2^2
0×2^3
0×2^4
1×2^5

while(n>0){
rem = n%10
sum = sum+rem×mul
n = n/10
mul = mul×2
}

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int mul = 1;
        int sum = 0;
        while (n > 0) {
            int rem = n % des;
            sum = sum + rem * mul;
            n = n / des;
            mul = mul * src;
        }
        System.out.println(sum);

    }
```

src des
10

3214 5 9 10 new num

Data Types

primitive

non primitive

byte b = 5

- byte → 8 bit → 1 byte
- short → 16 bit → 2 byte
- int → 32 bit → 4 byte
- long → 64 bit → 8 byte
- float → 32 bit → 4 byte
- double → 64 bit → 8 byte
- char → 2 byte → 16 bit
- boolean → True/False → 1 bit

Range

floating

8 bit

Sign bit
M.SB

-ve 1 1 1 1 1 1 1 1
+ve 0 1 1 1 1 1 1 1

$2^8 = 256$

[-127, 127] → 255

~~1 000 0000 → -0~~
~~0 000 0000 → +0~~

~~1's complement~~
~~0000~~
~~1111~~

0 0 0 1 0 1 0 0 → x
1 1 1 0 1 0 1 1 → -x

2's complement

0 1 1 0 1 1 0 1 → x
1 0 0 1 0 0 1 0
0 0 0 0 0 0 0 1
1 0 0 1 0 0 1 1 → -x

(-5) -8 bit

-ve 1 0 0 0 0 0 0 0
+ve 0 1 1 1 1 1 1 1

[-128, 127]

1 0 0 0 0 0 0 0
0 1 1 1 1 1 1 1
+ 0 0 0 0 0 0 0 1
1 0 0 0 0 0 0 0

2%2=0
2/2=1

byte	8 bit	-128 to 127	-2^7 to 2^7-1
Short	16 bit		-2^{15} to $2^{15}-1$
int	32 bit		-2^{31} to $2^{31}-1$
long	64 bit		-2^{63} to $2^{63}-1$

2^{31} ---- $2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$

-2^{31} to $2^{31}-1$ → 10^9
-2^{63} to $2^{63}-1$ → 10^{18}
long long

byte b = 5

0 0 0 0 0 1 0 1 ✓

byte b = -5

0 0 0 0 0 1 0 1
1 1 1 1 1 0 1 0
1
1 1 1 1 1 0 1 1 → -5

-ve

```
byte b = (byte)(428);
```

X 1 1 0 1 0 1 1 0 0

1 0 1 0 1 1 0 0
0 1 0 1 0 0 1 1
1
0 1 0 1 0 1 0 0
-84

1 0 0 1 0 1 1 0 0

```
byte b = (byte)(300);
```

0 0 1 0 1 1 0 0
+44

00000

32 zero
24 zero → 00000101

int i = 5

1 1 1 1 1 0 1 1

16 → 2^{16}

256

char
ASCII
Unicode

97 ← a
98 ← b
99 ← c
z

65 A
66 B
67 C
Z

'0' 48
'1' 49
'2' 50
51
'9'

26 → a to z
26 → A to Z
52
5
21
19 c
2
5v

```
for (byte i = 0; i < 128; i++) {
            System.out.println(i);
        }
```

byte (i+1)
~~byte (127+1)~~
-128

0
1
126
127
-128